

TECHNICAL BRIEF

Resolving Mobile Money (MoMo) Prompt Failures in High-Volume POS Environments

Target Platform	Paystack API (Mobile Money Core)	Issue Stage	OTP / USSD PUSH PROMPT
Prepared For	Systems Architecture / Development Team	Status	RESOLUTION READY

1. Executive Summary & Root Cause

The deployment of random email strings to bypass Paystack's single-IP rate limits successfully decoupled the IP blocking mechanism but systematically triggered downstream security and transaction failure layers. Specifically, transactions are failing at the critical **OTP / MoMo prompt stage**.

When transactions originate from a localized Point-of-Sale (POS) infrastructure, a massive volume of API requests hits Paystack from a singular IP address. While randomizing email strings masks the requests from basic edge rate limits, it fails the downstream **Fraud and Velocity Risk Assessment Engine**.

The Handshake Disruption Mechanism

- Telecom Provider Anti-Spam suppression:** Mobile Network Operators (MNOs like MTN, Telecel, AT) monitor incoming push requests. If an unverified domain fires rapid requests using randomly generated strings, the MNO suppresses the network-level interactive USSD/PIN popup to protect the subscriber from phishing or spam.
- First-Time Profile Penalty:** Paystack tracks customer reputation via the email parameter as a unique identifier. Constant randomization forces the gateway to evaluate every customer as a brand-new entity with zero history, triggering the highest possible friction and risk scoring.

Core Technical Reality: You do not need to collect genuine customer emails at a fast-moving physical terminal, but the values supplied to Paystack must be structured, predictable, and tied directly to an authenticated merchant domain.

2. Solution Implementation Architecture

To permanently eliminate the prompt drop-offs, the payload generation module must shift from **Random Generation** to **Deterministic Formatting**.

Step 1: Implement Deterministic Customer Data Profiles

Reconfigure your backend server (e.g., your Django, Flask, or Node.js API client) to compute a predictable email structure derived directly from the customer's payment telephone number. This grounds the customer profile on an authorized, recognized domain name.

```
// Corrected Paystack API Initialization Payload Structure
{
  "amount": 50000,                // Value in minor digits (e.g., 500.00 GHS)
  "currency": "GHS",
  "email": "momo_233245551234@yourposdomain.com", // Deterministic & Domain-Bound
  "mobile_money": {
    "phone": "0245551234",
    "provider": "mtn"
  },
  "metadata": {
    "device_type": "pos_terminal",
    "terminal_identifier": "accra_hub_01",
    "is_pos_checkout": "true"
  }
}
```

Step 2: Forward Original Client Fingerprints

If your physical POS terminal is an intelligent client or a web application capable of routing its own connection, ensure that your backend proxy application does not swallow the terminal's true local network IP address. Pass the terminal's IP downstream to keep individual checkouts decoupled from the main server's IP signature.

3. Merchant Whitelisting Protocol

Because localized POS networks inherently violate standard distributed web traffic patterns, automated firewalls will always require adjustments. You must explicitly register your technical setup with Paystack Merchant Support.

Recommended Support Ticket Template:

Subject: POS Integration Optimization Request: Whitelisting Single-Source MoMo Gateway Traffic

Hi Support Team,

We are running a custom high-volume Point-of-Sale (POS) system integrated via Paystack Mobile Money APIs. Because multiple terminal requests naturally route through our secure central server IP (**[Insert Your Server Static IP Here]**), we are restructuring our integrations to ensure absolute compliance with your anti-fraud standards.

We have updated our payload engine to map customer sessions using a predictable, domain-verified schema: `momo_[phone_number]@yourposdomain.com`.

Kindly update our merchant account flags to acknowledge our centralized single-source POS setup so that our high-frequency local MoMo transactions are not throttled or dropped during the telecom OTP/USSD push phase.

4. Summary of Verification Steps

1. **Audit Paystack Logs:** Look up recent failed requests on your Paystack Dashboard. Verify if the error timeline marks the transaction as *"Declined by fraud engine"*.
2. **Deploy Schema Change:** Push the phone-based deterministic email patch to production.
3. **Perform Live Tests:** Verify that the interactive PIN/USSD prompt drops immediately onto the target handset without network delay.